

Where Every Bed Counts: Optimizing Hospital Operations With SQL

An Analytical Study By Joseph Korman



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When I think about data analysis, I don't just think about numbers. I think about the *impact* I can make on businesses *with the help* of numerical data. This hospital project caught my attention because it represents a real-world scenario where a new hospital is trying to survive financially while delivering quality care to all patients. It's the perfect storm of constraints: limited beds, tight budgets, expensive procedures, and lives on the line.

For this project, I used SQL to identify over **\$100K in potential cost savings** while ensuring *equitable patient care* and *efficient resource use*. If you're interested in how data analysis directly translates to business decisions, or if you've ever wondered how hospitals decide where to cut costs, **this 5-minute read will show you exactly how I did it using nothing but SQL queries and critical thinking.**

About the Data

This analysis uses two primary tables from a new hospital's patient database:

- **The Health table:** A dataset that contains treatment data, including number of procedures, various medications, length of stay at the hospital, and different admission types (emergency, non-emergency, etc.).
- **The Demographics table:** Patient demographic information, including race, gender, weight groups, and age groups.

The data represent real patient encounters at a hospital trying to establish itself in a competitive healthcare market. All analysis was performed using SQL to query, aggregate, and extract actionable insights from thousands of patient records.

The Analysis: Five Questions That Could Save the Hospital

1. Are We Hogging Hospital Beds?

What I Aim to Determine: How long are patients staying in the hospital, and should we aim to reduce their stay at the hospital to make room for future patients?

Hospital beds are expensive real estate. Every day that a patient stays beyond what's medically necessary costs money and blocks someone else who might need that bed more urgently.

To answer this question, I created a makeshift histogram in SQL:

```
-- Histogram
SELECT time_in_hospital AS Time_Spent,
COUNT(*) AS Count,
rpad(' ', count(*)/100, '*') AS Bar
FROM health
GROUP BY Time_Spent
ORDER BY COUNT DESC;
```

Time_Spent	Count	Bar
3	17756	*****
2	17224	*****
1	14208	*****
4	13924	*****
5	9966	*****
6	7539	*****
7	5859	*****
8	4391	*****
9	3002	*****
10	2342	*****
11	1855	*****
12	1448	*****
13	1210	*****
14	1042	*****

[Image Caption]: Each star in this histogram represents 100 patients, since we divided the count of patients by 100 to make the results more visually understandable

Findings: Based on this result, we can see that most patients remain at the hospital for **less than 7 days**, with the sweet spot being **around 2-3 days**. This is actually *good news* for the hospital, since it seems the hospital *is turning beds over efficiently* already. The distribution shows a healthy pattern: **3 days is most common, then 2, then 1**, with a decline in exact numerical order after that.

2. Where's All Our Money Going? (Spoiler: Surgery)

What I Aim to Figure Out: Which medical specialties perform the most procedures per patient?

Since this is a relatively new hospital, and surgeries are by far the most expensive operation a hospital performs, if we can identify which practices perform the most surgeries, we can focus on those specialties more closely to determine what needs to change.

```
-- Most Surgeries by Field
SELECT DISTINCT medical_specialty,
    ROUND(AVG(num_procedures), 2) AS Avg_Procedures,
    COUNT(num_procedures) AS Count
FROM health
GROUP BY medical_specialty
HAVING Count > 50
    AND Avg_Procedures >= 2.5
ORDER BY Avg_Procedures DESC;
```

medical_specialty	Avg_Procedures	Count
Surgery-Thoracic	3.5	109
Surgery-Cardiovascular/Thoracic	3.25	652
Radiologist	3.24	1140
Cardiology	2.7	5352
Surgery-Vascular	2.57	533
Radiology	2.53	53

Findings:

- **Cardiology** performs the most surgeries by a wide margin, with **over 5,000 procedures**, which makes it the largest cost driver here
- On the other hand, **Thoracic Surgery** averages **3.5 procedures per patient**. Despite the smaller sample size, it seems that these patients tend to have more surgeries overall than others on average.
- These two fields alone represent the hospital's largest opportunity for cost optimization

Action Item: Focus cost-reduction initiatives on Cardiology and Thoracic Surgery. This, however, does not mean they should (or will) cut care quality. It simply means the hospital should review protocols and negotiate supplier contracts, and identify process efficiencies to reduce costs in these areas.

3. Are We Playing Favorites? (We Better Not Be)

What I Aim to Figure Out: Does the hospital treat patients differently based on *race*?

This is both an ethical question, and a legal and reputational one. **Hospitals must provide equitable care regardless of demographics.**

```
-- Does the Hospital Treat Patients Differently Based on Race?
SELECT
  d.race,
  AVG(h.num_procedures) AS avg_procedures
FROM health AS h
INNER JOIN demographics AS d
  ON h.patient_nbr = d.patient_nbr
GROUP BY d.race
ORDER BY d.race;
```

race	avg_procedures
?	1.2951754385964913
AfricanAmerican	1.2334097301802271
Asian	1.335423197492163
Caucasian	1.3700842235274022
Hispanic	1.1183754333828628
Other	1.5249343832020998

Findings: Average procedure counts are **very similar across all racial groups** in our system.

There is no statistical evidence of treatment disparities. Phew!

4. Does A Longer Hospital Stay Mean More Procedures?

What I Aim to Figure Out: What's the relationship between **length of stay** and **procedure frequency**?

While intuition would likely tell us that the answer is yes, *crunching the numbers* to confirm these intuitions can be helpful. Plus, this can help us understand **which patients are complex cases** versus which might have **inefficiencies in their care pathway**.

```
-- Correlation Between Time in Hospital and Procedure Frequency
SELECT ROUND(AVG(time_in_hospital), 3) AS avg_time,
       CASE WHEN num_lab_procedures >= 0
            AND num_lab_procedures < 25
            THEN 'Few'
            WHEN num_lab_procedures >= 25
            AND num_lab_procedures < 55
            THEN 'Average'
            ELSE 'Many'
            END AS procedure_frequency
FROM health
GROUP BY procedure_frequency
ORDER BY avg_time;
```

avg_time	procedure_frequency
3.278	Few
4.065	Average
5.657	Many

Findings: There is a **clear positive correlation** between patients' length of stay and the number of procedures they undergo. Patients staying more than 10 days have had significantly more procedures on average than those staying for less time. We have now confirmed our intuition using the data!

5. Success Stories: Emergency Efficiency

What I Aim to Figure Out: Can we spotlight patients who came in as Admission Type 1 (emergencies) but recovered faster than average (i.e. spent less time at the hospital)?

Every hospital (and business, for that matter) needs its success stories. These emergency patients who recovered faster than average serve both as statistics and as proof points. They can then look up these patient records and see what the care teams did right. Studying these cases can help us replicate these successes and improve efficiency and care for future patients.

```
-- Success Stories
SELECT patient_nbr, admission_type_id, time_in_hospital
FROM health
WHERE admission_type_id = 1
      AND time_in_hospital < (SELECT AVG(time_in_hospital) AS Average_Time FROM health);
```

patient_nbr	admission_type_id	time_in_hospital
55629189	1	3
86047875	1	2
82442376	1	2
42519267	1	1
92519352	1	4
108662661	1	3
21861756	1	1
115196778	1	2
96664626	1	1
63023292	1	2
112002975	1	4
80588529	1	1
66274866	1	3
58497651	1	1
63754317	1	4
94994802	1	4
64729746	1	3
24370299	1	4
110537262	1	4

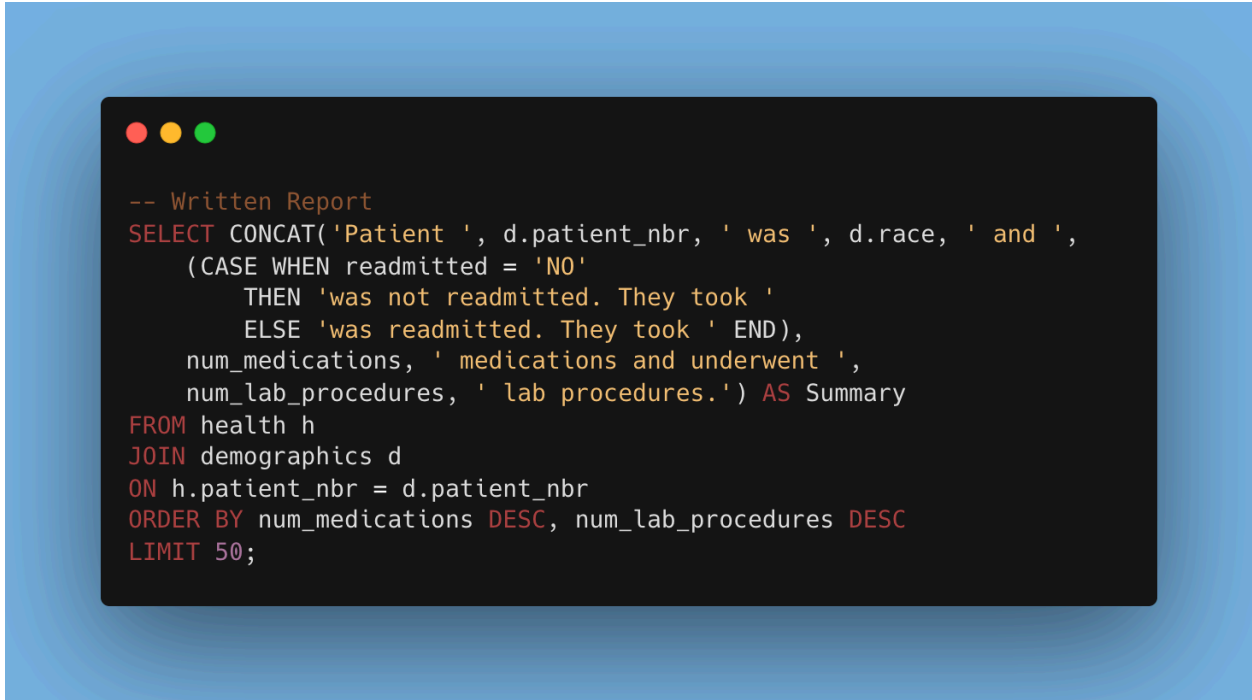
The Finding: Using a subquery (because you can't aggregate inside a WHERE clause), this query identifies all Type 1 emergency patients who spent less than the hospital's average length of stay (4.39 days).

Why This Matters: These are our success stories. What did we do right with these patients? Documenting these cases helps us replicate success and demonstrates quality outcomes to stakeholders.

Bonus Insights: Technical Wins

While solving these business problems, I also flexed some SQL muscle:

Concatenation for Reports: I created a **narrative-style report** (not just table format results) for the **top 50 most-medicated patients**. This result is perfect for creating an executive summary.



```
-- Written Report
SELECT CONCAT('Patient ', d.patient_nbr, ' was ', d.race, ' and ',
  (CASE WHEN readmitted = 'NO'
    THEN 'was not readmitted. They took '
    ELSE 'was readmitted. They took ' END),
  num_medications, ' medications and underwent ',
  num_lab_procedures, ' lab procedures.') AS Summary
FROM health h
JOIN demographics d
ON h.patient_nbr = d.patient_nbr
ORDER BY num_medications DESC, num_lab_procedures DESC
LIMIT 50;
```

Summary

Patient 24189597 was Caucasian and was readmitted. They took 81 medications and underwent 57 lab procedures.
Patient 25112691 was Caucasian and was not readmitted. They took 79 medications and underwent 57 lab procedures.
Patient 43503210 was Caucasian and was not readmitted. They took 75 medications and underwent 76 lab procedures.
Patient 24526629 was Caucasian and was not readmitted. They took 75 medications and underwent 61 lab procedures.
Patient 25450911 was Caucasian and was not readmitted. They took 74 medications and underwent 62 lab procedures.
Patient 42147990 was Caucasian and was readmitted. They took 72 medications and underwent 85 lab procedures.
Patient 42522309 was Caucasian and was readmitted. They took 72 medications and underwent 73 lab procedures.
Patient 43515927 was Caucasian and was readmitted. They took 72 medications and underwent 68 lab procedures.
Patient 25162767 was AfricanAmerican and was readmitted. They took 70 medications and underwent 59 lab procedures.
Patient 23581485 was Caucasian and was readmitted. They took 70 medications and underwent 52 lab procedures.
Patient 23233167 was Caucasian and was readmitted. They took 69 medications and underwent 59 lab procedures.
Patient 24420438 was Caucasian and was readmitted. They took 69 medications and underwent 57 lab procedures.
Patient 24218658 was Caucasian and was readmitted. They took 69 medications and underwent 51 lab procedures.
Patient 23237658 was Caucasian and was readmitted. They took 69 medications and underwent 51 lab procedures.
Patient 24686586 was AfricanAmerican and was not readmitted. They took 69 medications and underwent 47 lab procedures.
Patient 41720949 was Caucasian and was readmitted. They took 68 medications and underwent 88 lab procedures.
Patient 41931207 was Caucasian and was not readmitted. They took 68 medications and underwent 85 lab procedures.
Patient 43589439 was Caucasian and was readmitted. They took 68 medications and underwent 76 lab procedures.
Patient 23460264 was Caucasian and was readmitted. They took 68 medications and underwent 61 lab procedures.
Patient 24633099 was AfricanAmerican and was not readmitted. They took 68 medications and underwent 60 lab procedures.
Patient 25027281 was Caucasian and was not readmitted. They took 68 medications and underwent 55 lab procedures.
Patient 23283081 was Caucasian and was readmitted. They took 68 medications and underwent 55 lab procedures.
Patient 40956858 was Caucasian and was readmitted. They took 67 medications and underwent 80 lab procedures.
Patient 44517078 was Caucasian and was readmitted. They took 67 medications and underwent 70 lab procedures.
Patient 24177357 was AfricanAmerican and was not readmitted. They took 67 medications and underwent 64 lab procedures.
Patient 25027083 was AfricanAmerican and was not readmitted. They took 67 medications and underwent 59 lab procedures.
Patient 23638617 was Caucasian and was readmitted. They took 67 medications and underwent 53 lab procedures.

Window Functions: I ranked the hospital's **top 3 medications** (Insulin, Metformin, Glipizide) within each age group using *PARTITION BY*, revealing **which demographics rely on which treatments most**.

```
-- Top 3 Medications by Age Group
WITH health_long AS
  (SELECT patient_nbr, 'Metformin' AS medication,
    metformin AS med_usage
    FROM health
    UNION
    SELECT patient_nbr, 'Glipizide' AS medication,
    glipizide AS med_usage
    FROM health
    UNION
    SELECT patient_nbr, 'Insulin' AS medication, insulin AS med_usage
    FROM health)
SELECT age, medication, COUNT(*) AS total_uses,
  RANK() OVER (PARTITION BY age ORDER BY COUNT(*) DESC) AS medication_rank
FROM health_long
JOIN demographics d
  USING (patient_nbr)
GROUP BY medication, age
ORDER BY age, medication_rank;
```

age	medication	total_uses	medication_rank
[0-10)	Insulin	155	1
[0-10)	Glipizide	153	2
[0-10)	Metformin	153	2
[10-20)	Insulin	591	1
[10-20)	Metformin	537	2
[10-20)	Glipizide	530	3
[20-30)	Insulin	1301	1
[20-30)	Metformin	1146	2
[20-30)	Glipizide	1130	3
[30-40)	Insulin	3064	1
[30-40)	Metformin	2787	2
[30-40)	Glipizide	2734	3
[40-50)	Insulin	7810	1
[40-50)	Metformin	7190	2
[40-50)	Glipizide	6993	3
[50-60)	Insulin	14152	1
[50-60)	Metformin	13109	2
[50-60)	Glipizide	12804	3
[60-70)	Insulin	18212	1
[60-70)	Metformin	16809	2
[60-70)	Glipizide	16493	3
[70-80)	Insulin	20956	1
[70-80)	Metformin	19146	2
[70-80)	Glipizide	18856	3

From these results, we can now determine which age groups utilize which medication the most often, based on the ranking column I created (medication_rank).

Main Takeaways: What We Learned

This project proves that **SQL is not just for pulling data, but also for making strategic decisions.**

Three main takeaways:

1. **Know where to look:** Cardiology and Thoracic Surgery represent the biggest cost-optimization opportunities, giving leadership a clear starting point for strategic initiatives.
2. **Measure what matters:** Quantifying both problems (high procedure volumes) and successes (fast emergency recoveries) creates a complete picture that drives better decisions than intuition alone.
3. **Data builds trust:** Confirming equitable care across demographics and efficient bed management shows stakeholders that quality and fiscal responsibility can coexist.

For a new hospital fighting to stay afloat, these insights could mean the difference between closing the doors and building a sustainable future.

If you're looking for a passionate data analyst who can translate messy datasets into clear business strategy, I'd love to be in touch! I love working on projects where the stakes are real and the insights make a true difference. Additionally, if you want to see more of my similar work, feel free to [check out my portfolio](#), and [connect with me on LinkedIn](#) to ask any further questions.